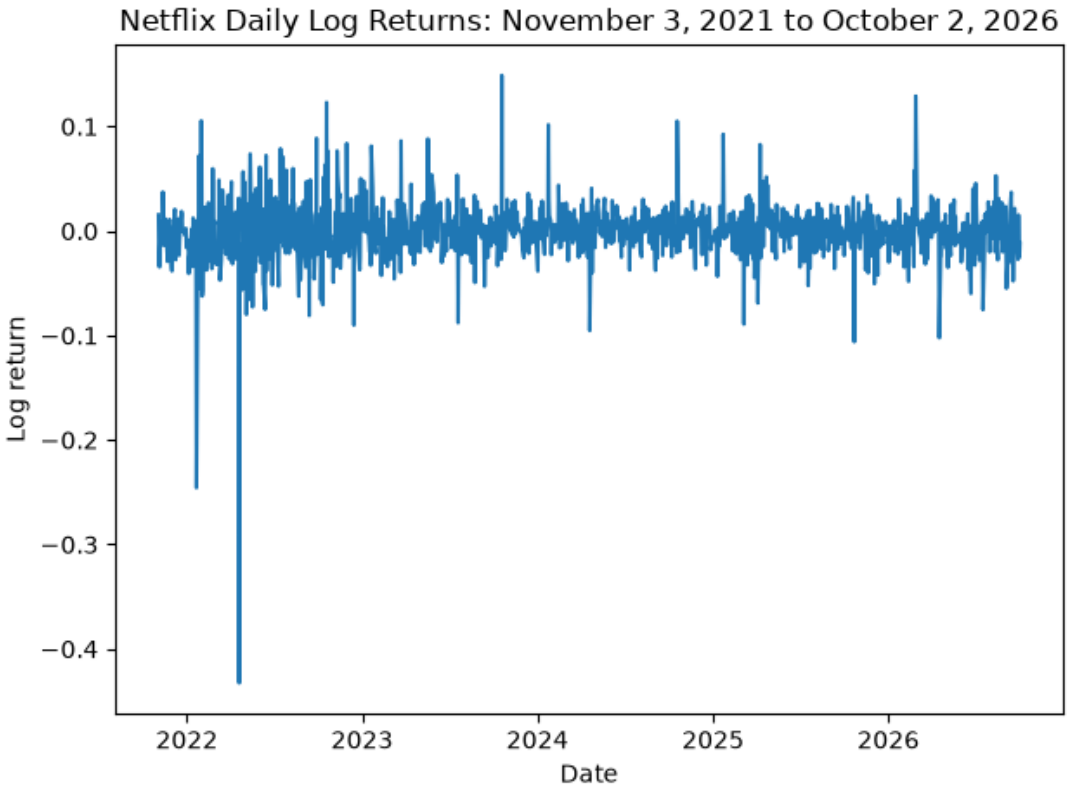
Writer: Kan Wai Kit | Analysis made on 2026-10-05 | Report written on 2026-10-21

Netflix Stock-Price Simulation Using the Heston Model

This project estimates Netflix's future price distribution using the Heston stochastic-volatility model.

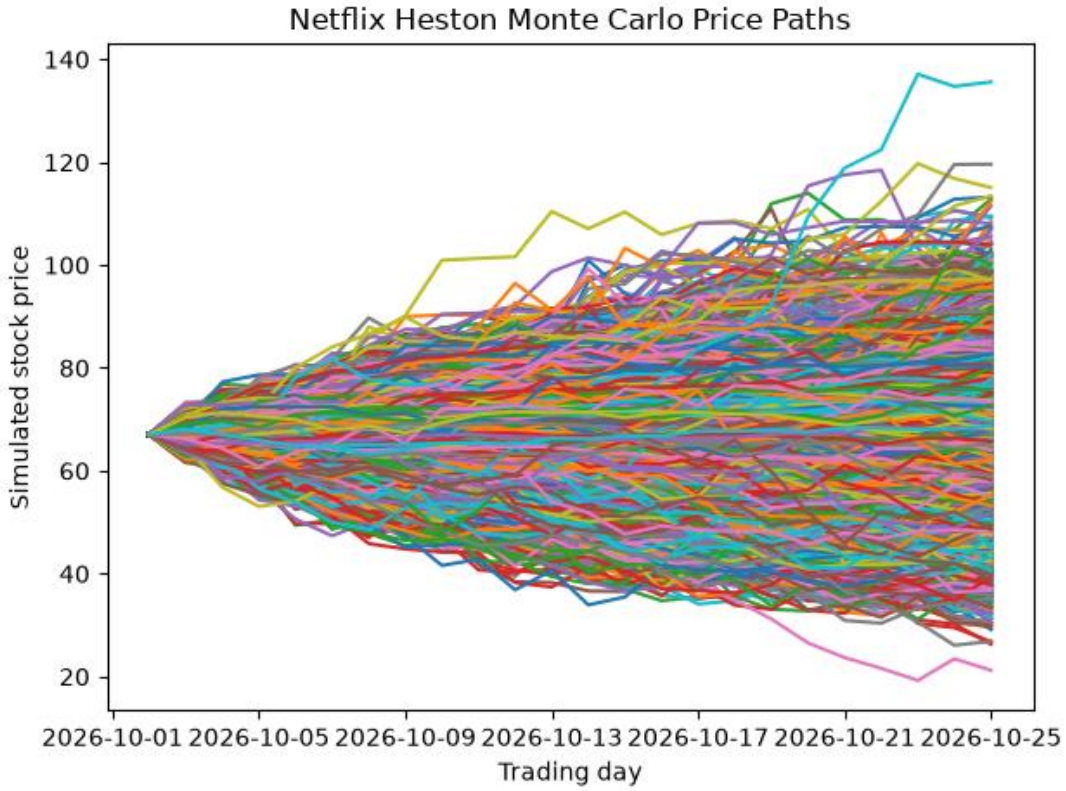
I estimated the model using historical returns and rolling variance, then simulated 50,000 price paths to assess expected return, profit probability, downside risk, Value at Risk, and Expected Shortfall.

Log return analysis:



Netflix's daily log returns fluctuate around zero, with changes in the dispersion of returns over time. Several large positive and negative observations indicate non-normal tail behaviour and time-varying volatility**. These characteristics support the use of a stochastic-volatility model**, although the extreme observations should be validated against adjusted price data before parameter estimation.

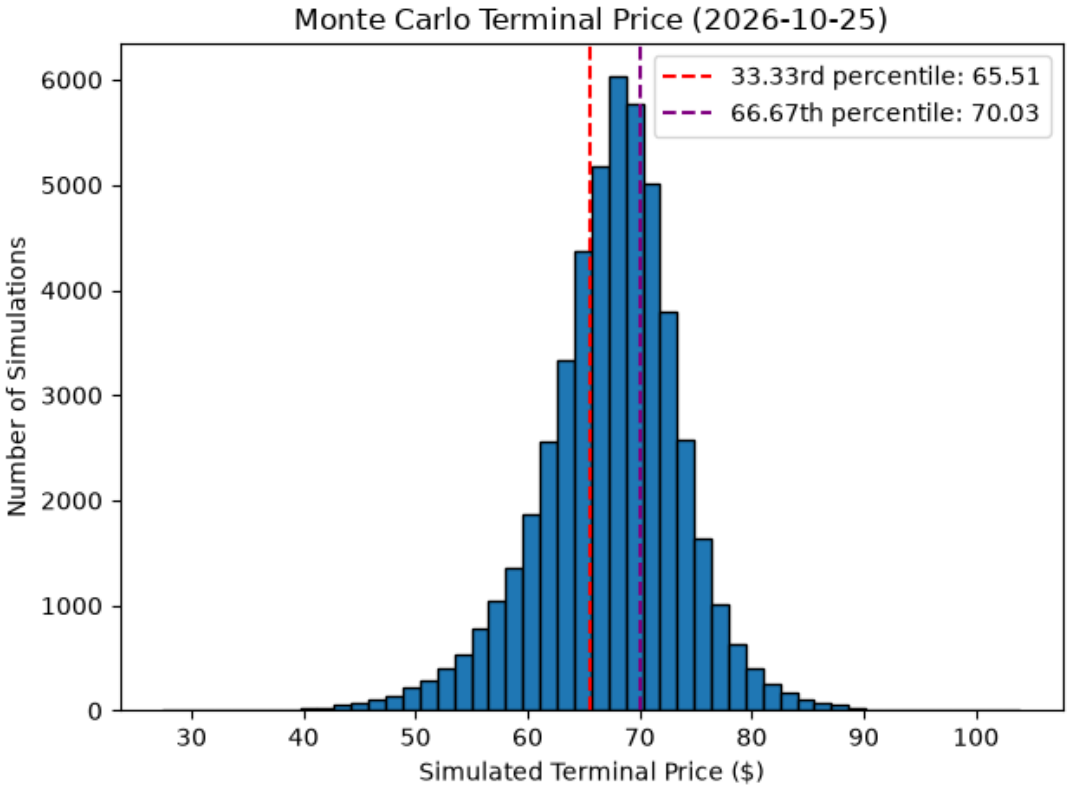
Simulation result:



Calculation output:

- Expected return: 0.4881%(simulated holding-period return from 2026-10-02 to 2026-10-23)
    - Beginning stock price: $67.06 (real price)
    - Mean ending stock price: $ 67.3873 (with 50,000 simulations)
- Median return = 1.2153%
- Probability of profit: 56.362%
- 95% Value at Risk: 16.3069% loss
- Average return among the worst 5% of outcomes: 22.5769% loss

Monte Carlo Terminal Price (2026-10-23):



Case-by-case price analysis:

Lower-third scenario: below $65.51

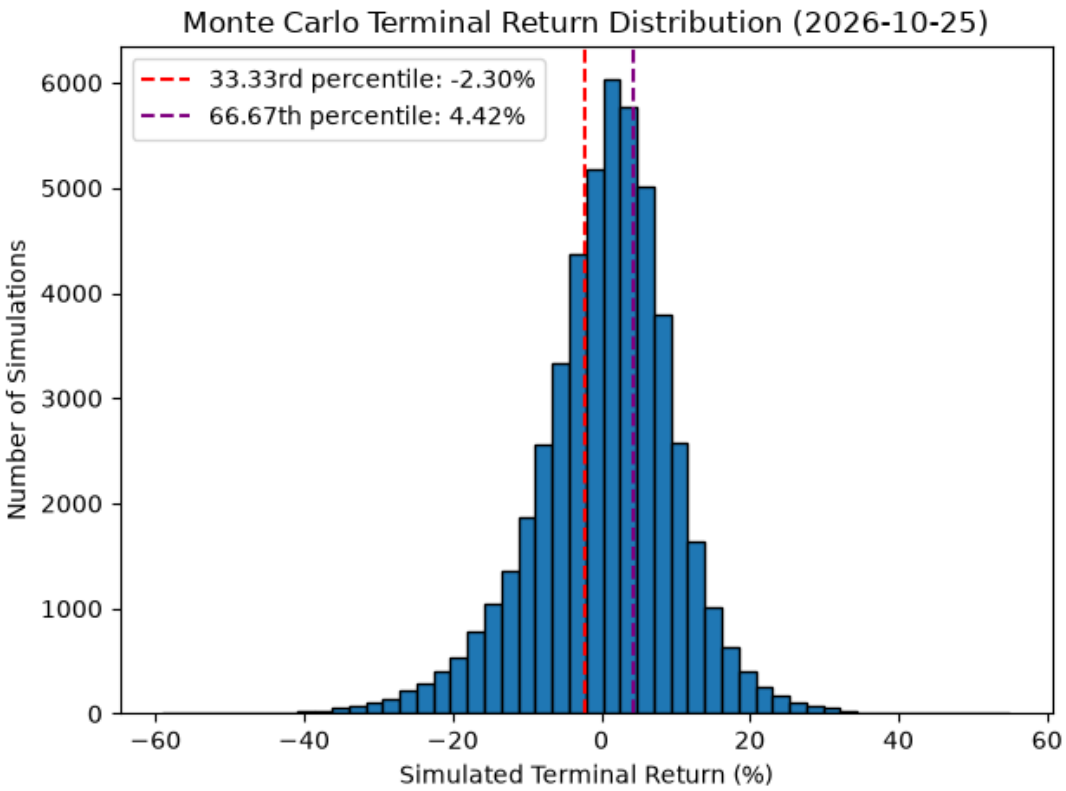
Middle-third scenario: $65.51 to $70.03

Upper-third scenario: above $70.03

Comment: The percentile thresholds divide the simulated terminal prices into three approximately equal groups. The middle-third range represents the central portion of the simulated distribution, but it should not be interpreted as containing the majority of potential outcomes.

In addition, if an investor buys the stock at the price of $67.06 (real price) on 2026-10-02, the investor can receive the following return:

Monte Carlo Terminal Return Distribution (2026-10-23):



Case-by-case return analysis:

Lower-third scenario: below -2.30%

Middle-third scenario: -2.30% to 4.42%

Upper-third scenario: above 4.42%

Conclusion:

The simulation indicates modest upside potential, with an expected return of **0.4881%**. However, the **Expected Shortfall of 22.5769%** indicates substantial downside risk, representing the average loss among the worst 5% of simulated outcomes. This highlights the importance of position sizing and downside-risk controls.

The results depend on historical data, estimated parameters and model assumptions. They represent model-generated scenarios rather than guaranteed forecasts or investment recommendations.